

Simple Kinetic Molecular Model of Matter

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Simple Kinetic Molecular Model of Matter
Booklet	Question Paper 1

Time allowed: 25 minutes

Score: /20

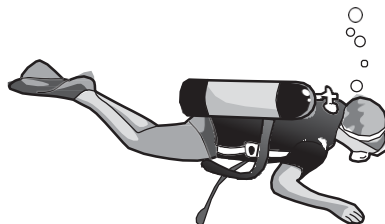
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

A diver under water uses breathing apparatus at a depth where the pressure is $1.25 \times 10^5 \text{ Pa}$.



A bubble of gas breathed out by the diver has a volume of 20 cm^3 when it is released. The bubble moves upwards to the surface of the water.

At the surface of the water, the atmospheric pressure is $1.00 \times 10^5 \text{ Pa}$.

The temperature of the water is the same at all depths.

What is the volume of this bubble when it reaches the surface?

- A 15 cm^3 B 16 cm^3 C 20 cm^3 D 25 cm^3

Question 2

- 14 Which statement about evaporation is correct?
- A. Evaporation causes the temperature of the remaining liquid to decrease.
 - B. Evaporation does not occur from a cold liquid near its freezing point.
 - C. Evaporation does not occur from a dense liquid, such as mercury.
 - D. Evaporation occurs from all parts of a liquid.

Question 3

Smoke particles, illuminated by a bright light, are seen through a microscope. They move about randomly.

What causes this motion?

- A attraction between the smoke particles and the molecules of the air
- B collisions between the smoke particles and the molecules of the air
- C evaporation of the faster-moving smoke particles
- D warming of the smoke particles by the lamp

Question 4

A liquid is at a temperature below its boiling point.

The liquid is then heated so that it becomes a gas at a temperature above its boiling point.

Which row correctly compares the liquid with the gas?

	average distance between the particles	average speed of the particles
A	greater in the liquid	greater in the liquid
B	greater in the liquid	smaller in the liquid
C	smaller in the liquid	greater in the liquid
D	smaller in the liquid	smaller in the liquid

Question 5

A beaker of liquid is left on a laboratory bench. There is an electric fan in the laboratory causing a draught over the liquid.

The liquid evaporates.

Which row shows two changes that will both cause the liquid to evaporate more quickly?

	change to surface area of the liquid	change to speed of fan
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

Question 6

What causes the random, zig-zag movement (Brownian motion) of smoke particles suspended in air?

- A air molecules colliding with smoke particles
- B convection currents as the hot smoke rises
- C smoke particles colliding with each other
- D smoke particles reacting with oxygen molecules in the air

Question 7

A sealed bottle of constant volume contains air.

The air in the bottle is heated by the Sun.

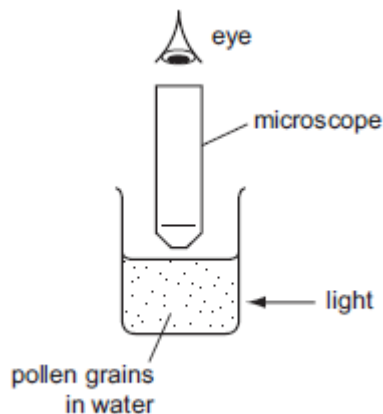
What is the effect on the average speed of the air molecules in the bottle, and the average distance between them?

	average speed of air molecules	average distance between air molecules
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

Question 8

Very small pollen grains are suspended in a beaker of water. A bright light shines from the side.

Small, bright dots of light are seen through a microscope. The dots move in rapidly changing, random directions.



What are the bright dots?

- A. pollen grains being hit by other pollen grains
- B. pollen grains being hit by water molecules
- C. water molecules being hit by other water molecules
- D. water molecules being hit by pollen grains

Question 9

A sealed gas cylinder is left outside on a hot, sunny day.

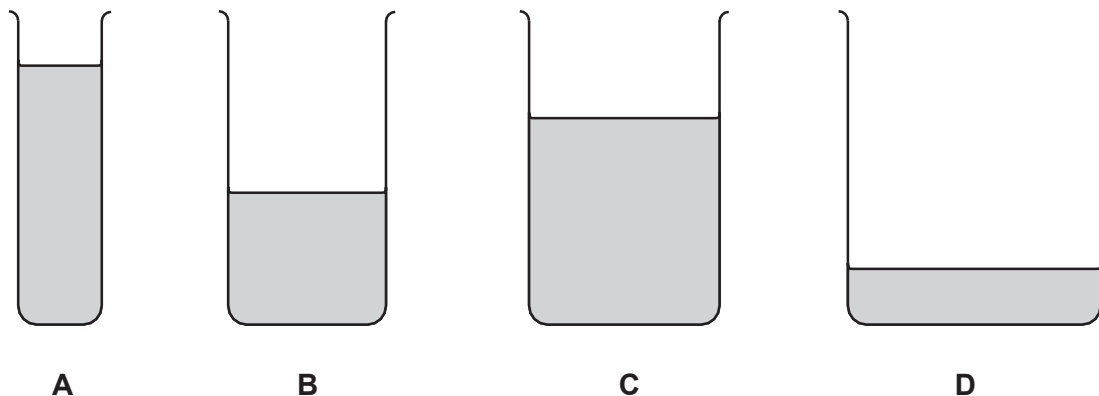
What happens to the average speed of the gas molecules and to the pressure of the gas in the cylinder as the temperature of the gas rises?

	average speed of gas molecules	pressure of gas in cylinder
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

Question 10

The diagram shows four beakers A, B, C and D. The beakers contain different amounts of the same liquid at the same temperature. The beakers are left next to each other on a laboratory bench overnight. The diagrams are all drawn to the same scale.

From which beaker does the largest quantity of liquid evaporate?



Question 11

Two states of matter are described as follows.

In state 1, the molecules are very far apart. They move about very quickly at random in straight lines until they hit something.

In state 2, the molecules are quite closely packed together. They move about at random. They do not have fixed positions.

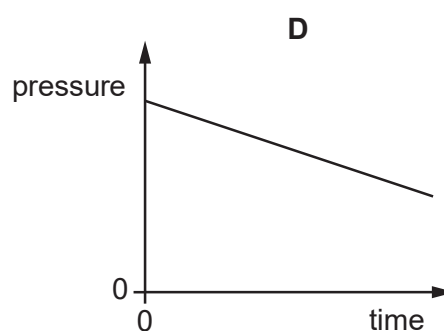
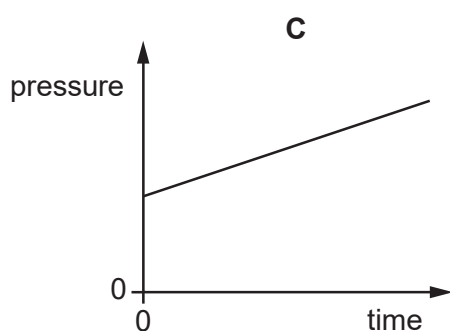
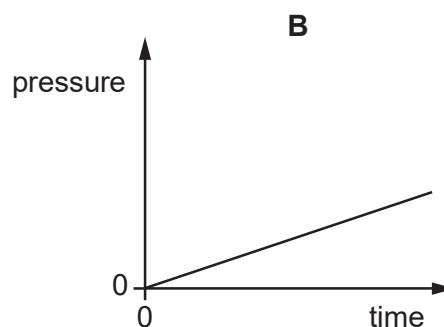
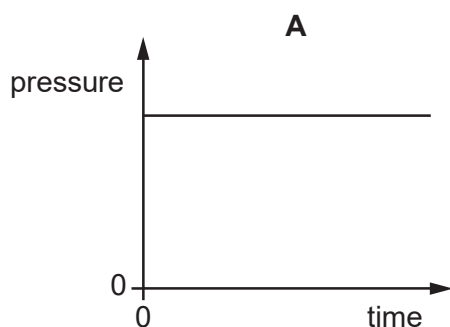
What is state 1 and what is state 2?

	state 1	state 2
A	gas	liquid
B	gas	solid
C	liquid	gas
D	solid	liquid

Question 12

The pressure of a fixed mass of gas in a cylinder is measured. The temperature of the gas in the cylinder is then slowly increased. The volume of the cylinder does not change.

Which graph shows the pressure of the gas during this process?



Question 13

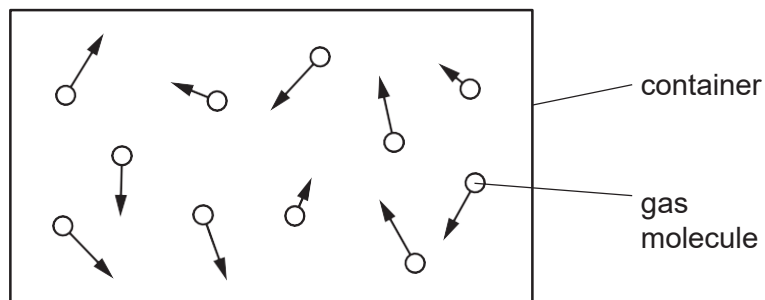
Puddles of rain water remain after a storm. The water in the puddles gradually evaporates.

How does the evaporation affect the temperature of the water remaining in the puddle, and how does it affect the average speed of the remaining water molecules in the puddle?

	temperature of water in puddle	average speed of water molecules in puddle
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

Question 14

The diagram represents moving gas molecules in a sealed container of fixed volume.



The temperature of the gas is now increased.

What happens to the pressure of the gas, and what happens to the speed of the gas molecules?

	pressure of gas	speed of molecules
A	increases	increases
B	increases	unchanged
C	unchanged	increases
D	unchanged	unchanged

Question 15

Small smoke particles suspended in air are viewed through a microscope.

The smoke particles move randomly.

What does this show?

- A. The air consists of fast-moving molecules.
- B. The pressure of the air is increasing.
- C. There are convection currents in the air.
- D. The temperature of the air is increasing.

Question 16

Molecules escape from a liquid during evaporation. The temperature of the remaining liquid changes.

Which molecules escape and how does the temperature change?

	molecules escaping	temperature of remaining liquid
A	least energetic	decreases
B	least energetic	increases
C	most energetic	decreases
D	most energetic	increases

Question 17

A cylinder of constant volume contains a fixed mass of gas. The gas is cooled.

What happens to the pressure of the gas and what happens to the kinetic energy of the gas molecules?

	pressure of gas	kinetic energy of molecules
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

Question 18

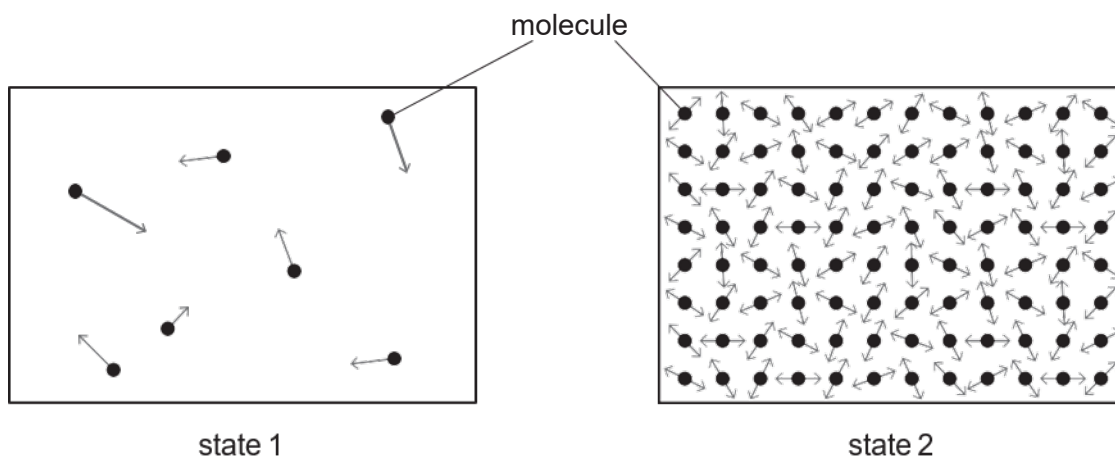
A swimmer feels cold after leaving warm water on a warm, windy day.

Why does she feel cold even though the air is warm?

- A. The less energetic water molecules on her skin escape quickly.
- B. The more energetic water molecules on her skin do not escape quickly.
- C. The water on her skin does not evaporate quickly enough to keep her warm.
- D. The water on her skin evaporates quickly and cools her skin.

Question 19

The diagrams represent the molecules in two different states of matter. The arrows show the motion of the molecules.



What is state 1, and what is state 2?

	state 1	state 2
A	gas	liquid
B	gas	solid
C	liquid	gas
D	liquid	solid

Question 20

A car tyre has a constant volume.

Why does the pressure of the air in the tyre increase when its temperature increases?

- A. The air molecules hit each other less often.
- B. The air molecules hit the inside of the tyre less often.
- C. The average speed of the air molecules in the tyre is greater.
- D. There are more air molecules in the tyre.

Simple Kinetic Molecular Model of Matter

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Simple Kinetic Molecular Model of Matter
Booklet	Question Paper 2

Time allowed: 23 minutes

Score: /18

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

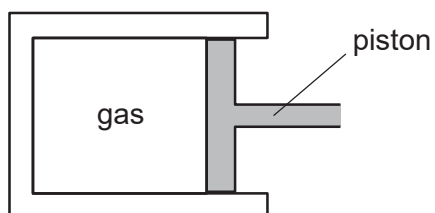
Extremely small pollen grains in water are viewed through a microscope. The grains are seen to move continually and randomly.

What is the reason for this random movement?

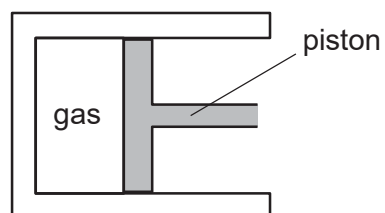
- A. The grains are moved by randomly moving water molecules.
- B. The grains are moved by random convection currents in the water.
- C. The grains are moved by random rays of light reflecting off them.
- D. The grains are moved by the random motion of their own atoms.

Question 2

A gas is compressed in a sealed cylinder by moving a piston.



initial position



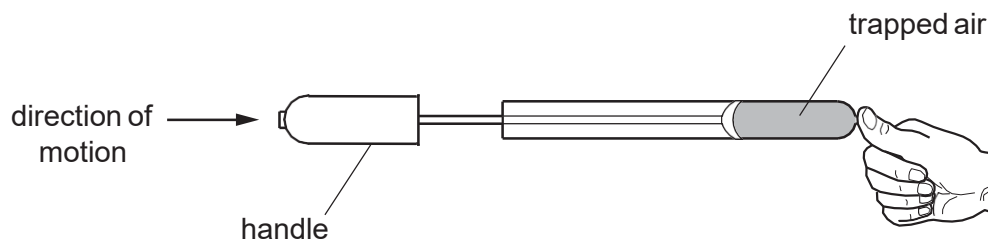
after gas has been compressed

Which row in the table states what happens to the density of the gas and to the pressure of the gas when it is compressed?

	density	pressure
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

Question 3

A student places his thumb firmly on the outlet of a bicycle pump, to stop the air coming out.



What happens to the pressure and what happens to the volume of the trapped air as the pump handle is pushed in?

	pressure	volume
A	decreases	decreases
B	decreases	remains the same
C	increases	decreases
D	increases	remains the same

Question 4

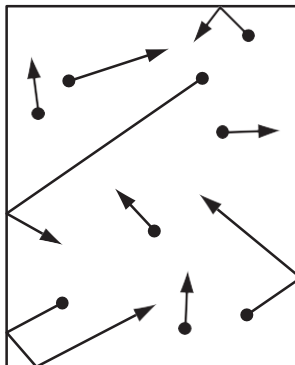
During evaporation, molecules escape rapidly from the surface of a liquid.

What happens to the average energy of the molecules of the remaining liquid and what happens to the temperature of the remaining liquid?

	average energy of remaining molecules	temperature of remaining liquid
A	decreases	decreases
B	decreases	increases
C	stays the same	decreases
D	stays the same	increases

Question 5

The diagram represents molecules of gas moving in a container.



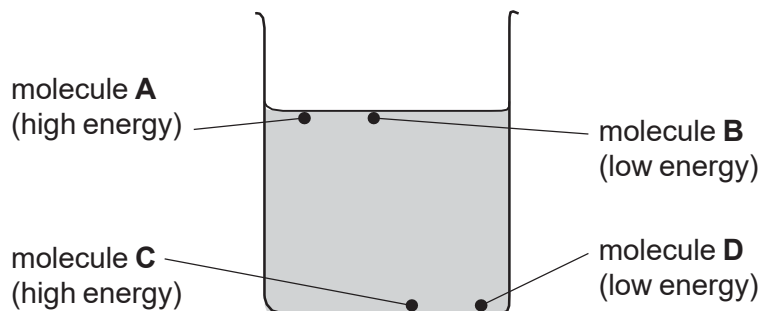
What happens to the gas molecules when the temperature of the gas increases?

- A They move more quickly.
- B They move more slowly.
- C They vibrate more quickly.
- D They vibrate more slowly.

Question 6

The diagram shows a beaker of water. Four molecules are labelled. The relative amount of energy of each molecule is shown.

Which molecule is most likely to escape from the liquid?

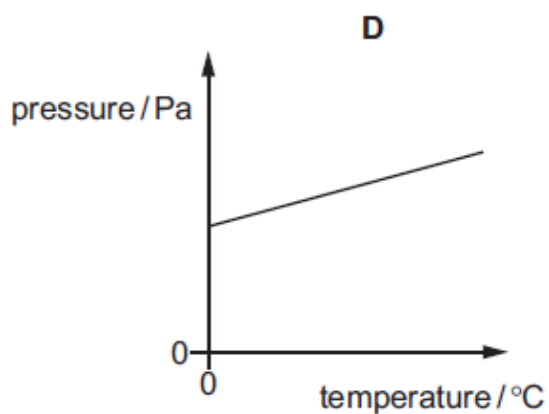
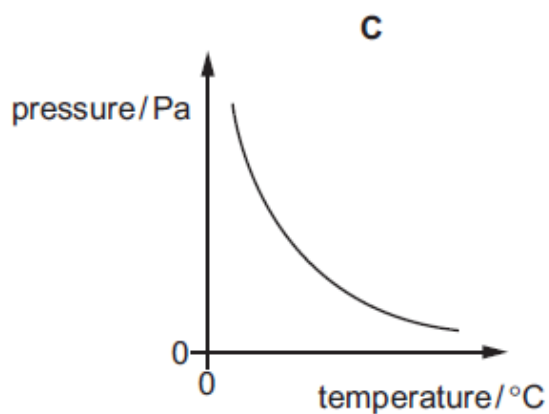
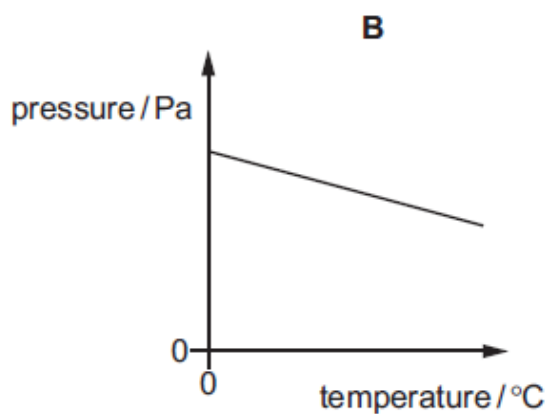
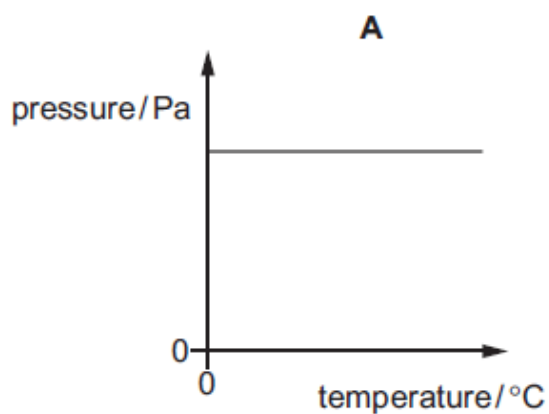


Question 7

Some gas is trapped in a container of fixed volume.

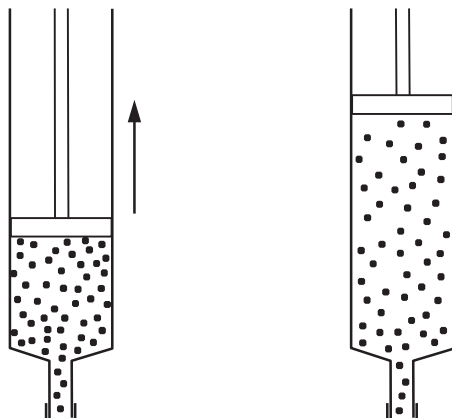
The temperature of the gas increases.

Which graph shows how the pressure of the gas changes with temperature?



Question 8

The volume of a gas in a sealed syringe is increased. The temperature of the gas does not change.



before change

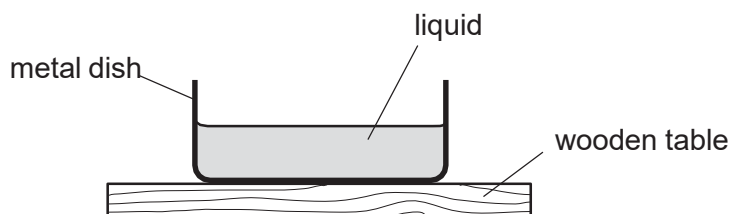
after change

After this change is made, what has happened to the gas molecules in the syringe?

- A. They move more quickly.
- B. They move more slowly.
- C. They hit the syringe walls less often.
- D. They hit the syringe walls more often.

Question 9

Some liquid is poured into a metal dish on a wooden table. The dish, the liquid, the table and the air around the dish are all at the same temperature.



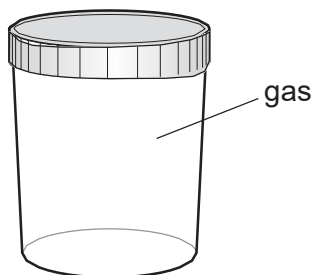
The temperature of the liquid now starts to decrease.

What could cause this temperature decrease?

- A. convection currents in the liquid
- B. conduction of heat through the metal dish
- C. evaporation of the liquid
- D. heat radiation from the liquid

Question 10

The diagram shows a sealed jar containing a gas.



Which statement about the gas in the jar is correct?

- A. The gas molecules collide with the inside of the jar more often as the temperature increases.
- B. The gas molecules move more slowly as the temperature increases.
- C. The pressure of the gas decreases as the temperature increases.
- D. The pressure of the gas is higher at the top of the jar than at the bottom of the jar.

Question 11

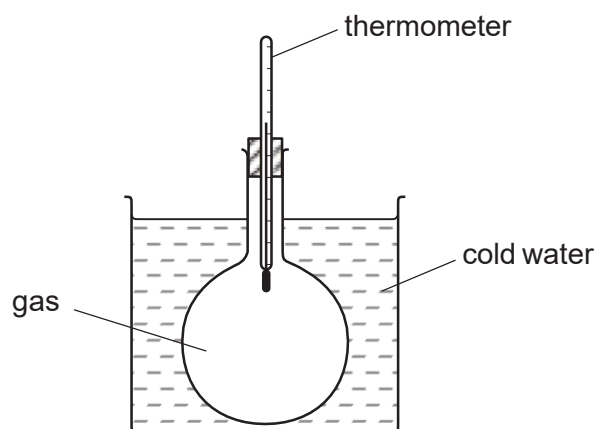
A block of ice cream is prevented from melting by wrapping it in newspaper soaked in water. The water evaporates from the newspaper.

Which molecules escape from the water and what happens to the average speed of the water molecules that remain in the newspaper?

	escaping molecules	average speed of the remaining water molecules
A	the less energetic ones	decreases
B	the less energetic ones	increases
C	the more energetic ones	decreases
D	the more energetic ones	increases

Question 12

A closed flask of gas is placed in a cold-water bath.



As the flask cools, the temperature of the gas decreases.

What happens to the molecules of the gas?

- A. They contract.
- B. They expand.
- C. They move more quickly.
- D. They move more slowly.

Question 13

Brownian motion is observed when looking at smoke particles in air using a microscope.

What causes the smoke particles to move at random?

- A. Smoke particles are hit by air molecules.
- B. Smoke particles are moved by convection currents in the air.
- C. Smoke particles have different weights and fall at different speeds.
- D. Smoke particles hit the walls of the container.

Question 14

The molecules of a substance become more closely packed and move more quickly.

What is happening to the substance?

- A. A gas is being heated and compressed.
- B. A gas is being heated and is expanding.
- C. A liquid is boiling.
- D. A liquid is evaporating at room temperature.

Question 15

When a liquid evaporates, some molecules escape from it and its temperature changes.

From where do the molecules escape and what is the effect on the temperature of the liquid?

	molecules escape from	temperature of liquid
A	all parts of the liquid	decreases
B	all parts of the liquid	increases
C	only the liquid surface	decreases
D	only the liquid surface	increases

Question 16

A sealed gas cylinder is left outside on a hot, sunny day.

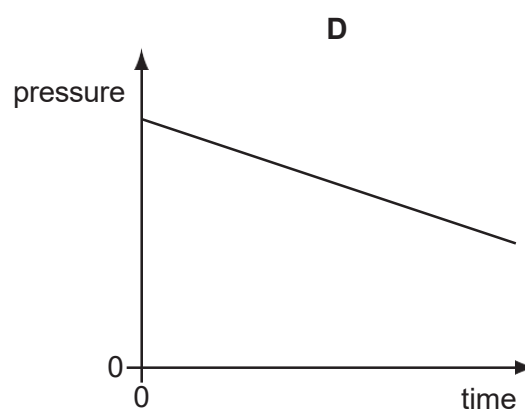
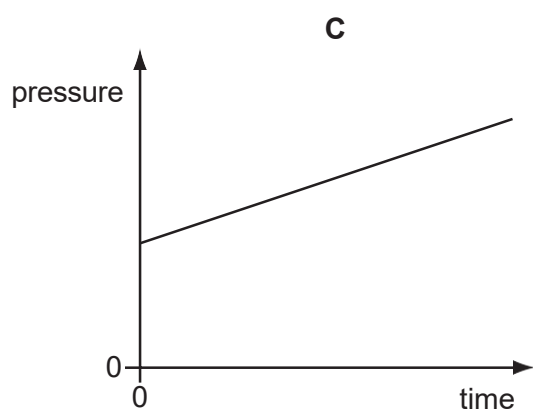
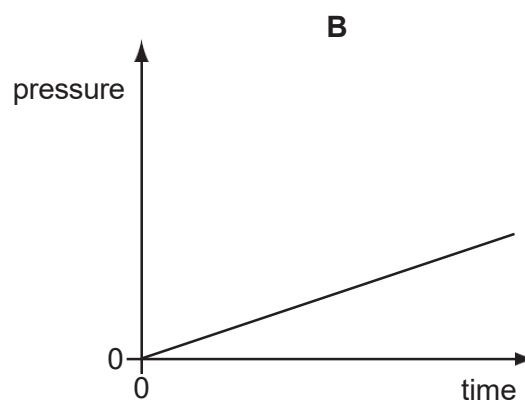
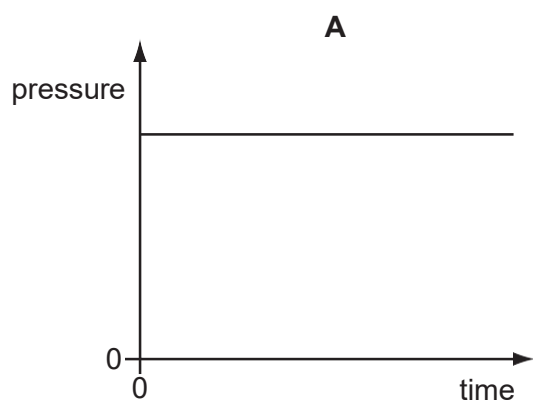
What happens to the average speed of the molecules and to the pressure of the gas in the cylinder as the temperature rises?

	average speed of the gas molecules	gas pressure
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

Question 17

The pressure of a fixed mass of gas in a cylinder is measured. The volume of the gas in the cylinder is then slowly decreased. The temperature of the gas does not change.

Which graph could show the change of pressure of the gas during this process?



Question 18

Evaporation occurs when molecules escape from a liquid surface into the air above it. During this process the temperature of the liquid falls.

Why does the temperature of the liquid fall?

- A The molecules in the vapour expand because the pressure is less.
- B The molecules left in the liquid have more space to move around.
- C The molecules move more slowly when they escape into the air.
- D The molecules with the highest energies escape into the air.

Thermal Properties & Temperature

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Thermal Properties & Temperature
Booklet	Question Paper 1

Time allowed: 24 minutes

Score: /19

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

A beaker contains 0.500 kg of water at a temperature of 3.0 °C. The beaker is heated, and the internal energy of the water increases by 21.0 kJ.

The specific heat capacity of water is 4200 J/(kg °C).

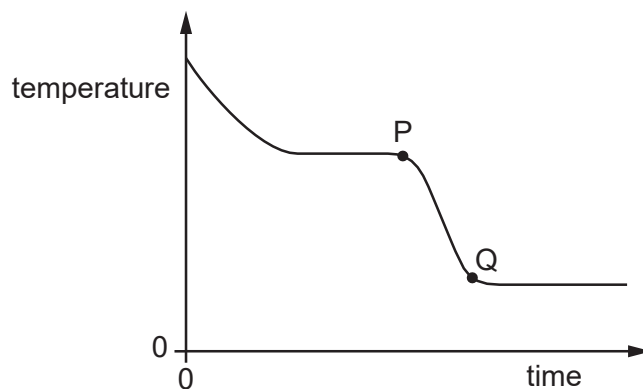
What is the temperature of the water after it has been heated?

- A 5.5 °C B 10.0 °C C 13.0 °C D 31.5 °C

Question 2

A substance loses thermal energy (heat) to the surroundings at a steady rate.

The graph shows how the temperature of the substance changes with time.

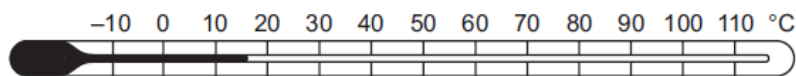


What could the portion PQ of the graph represent?

- A. gas condensing
- B. gas cooling
- C. liquid cooling
- D. liquid solidifying

Question 3

A student wishes to check the upper and the lower fixed points on a Celsius scale thermometer.



She has four beakers P, Q, R and S.

Beaker P contains a mixture of ice and salt.

Beaker Q contains a mixture of ice and water.

Beaker R contains boiling salt solution.

Beaker S contains boiling water.

Which two beakers should she use to check the fixed points?

- A P and R B P and S C Q and R D Q and S

Question 4

Which statement describes what happens as ice at 0 °C starts to melt to become water?

- A. Energy is absorbed and the temperature remains constant.
- B. Energy is absorbed and the temperature rises.
- C. Energy is released and the temperature remains constant.
- D. Energy is released and the temperature rises.

Question 5

5.0 g of water at 25 °C is dropped onto a large block of ice at 0 °C. The water cools to 0 °C and some of the ice melts.

Assume that all the energy lost by the water is gained by the ice.

What is the mass of ice that melts?

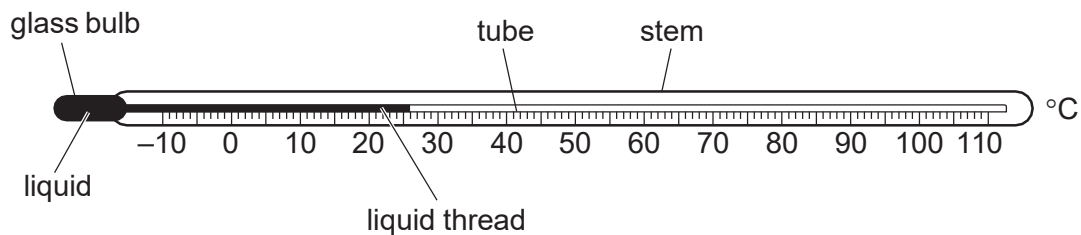
The specific heat capacity of water is 4.2 J/(g °C).

The specific latent heat of fusion of ice is 340 J/g.

- A 0.062g B 0.087g C 1.5g D 10g

Question 6

The diagram shows a liquid-in-glass thermometer.



How can the thermometer be made more sensitive?

- A. increase the internal diameter of the tube containing the liquid thread
- B. increase the internal volume of the glass bulb and the volume of the liquid
- C. increase the length of the tube and stem
- D. increase the thickness of the glass in the glass bulb

Question 7

In an experiment to measure specific heat capacity, a block of aluminium is heated and its rise in temperature is measured.

The amount of energy gained by the block is E . The mass of the block is m . The rise in temperature of the block is ΔT .

Which expression gives the specific heat capacity of aluminium?

A $\frac{m}{E\Delta T}$

B $\frac{m\Delta T}{E}$

C $\frac{E}{m\Delta T}$

D $\frac{E\Delta T}{m}$

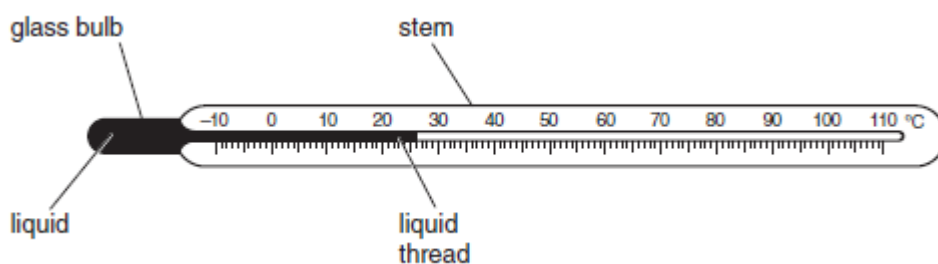
Question 8

Which line in the table shows the relative expansion of the three states of matter from the most expansion to the least expansion?

	most expansion			least expansion		
A	solids	>	liquids	>	gases	
B	solids	>	gases	>	liquids	
C	gases	>	liquids	>	solids	
D	gases	>	solids	>	liquids	

Question 9

The diagram shows a liquid-in-glass thermometer.



Which two features both affect the sensitivity of the thermometer?

- A. mass of liquid and diameter of liquid thread
- B. mass of liquid and length of stem
- C. thickness of glass bulb and diameter of liquid thread
- D. thickness of glass bulb and length of stem

Question 10

A student wishes to calculate the specific heat capacity of copper.

He has a block of copper and an electrical heater. He knows the power of the heater.

Which other apparatus does he need?

	balance	stop watch	thermometer
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = needed

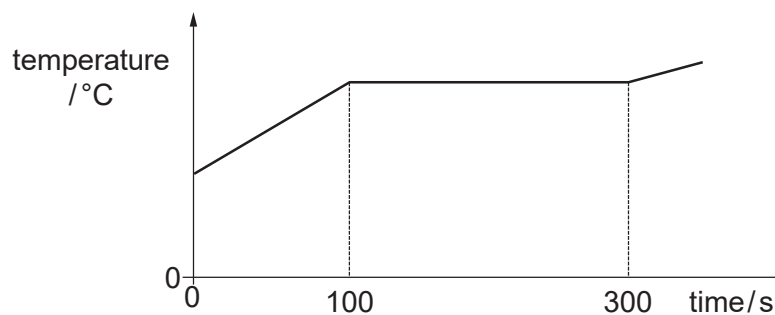
✗ = not needed

Question 11

A mass of 0.20 kg of a substance is initially solid.

It is heated at a steady rate of 500 W.

The graph shows how the temperature of the substance changes with time.

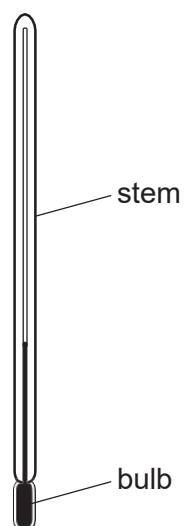


What is the specific latent heat of fusion of the substance?

- A 20 000 J/kg
- B 30 000 J/kg
- C 500 000 J/kg
- D 750 000 J/kg

Question 12

The thermometer in the diagram has no scale.



Where must the bulb be placed so that 0 °C can be marked on the stem?

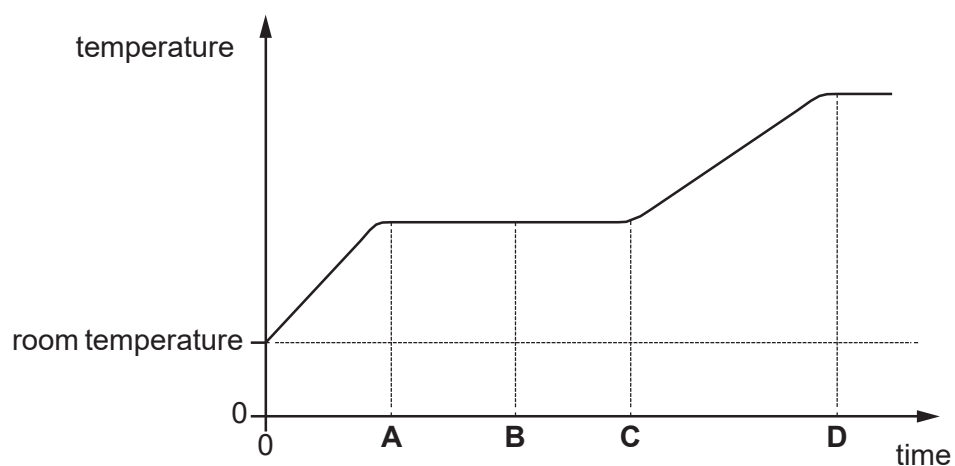
- A. in a freezer
- B. in pure boiling water
- C. in pure cold water
- D. in pure melting ice

Question 13

A solid is heated from room temperature.

The graph shows how its temperature changes with time as it is heated constantly.

At which time has it just become **completely** liquid?



Question 14

Two metal blocks X and Y are at room temperature. Each block is heated so that its temperature rises by 10°C .

The blocks are now allowed to cool back to room temperature.

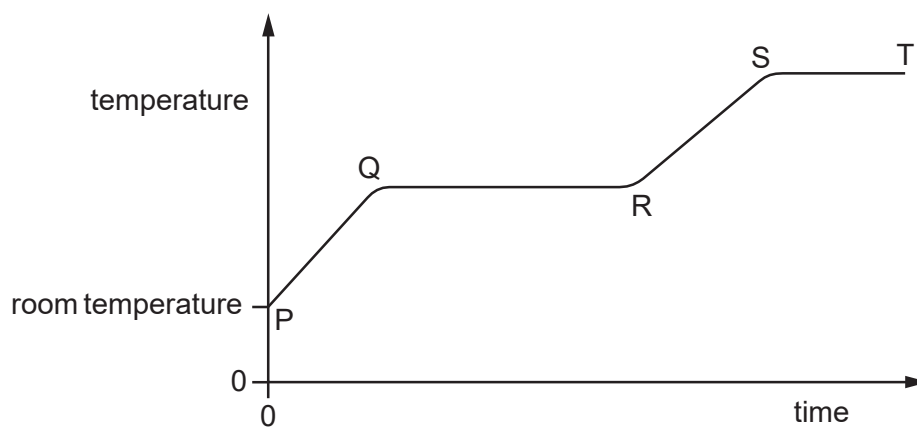
Block Y has a greater thermal capacity than block X.

Which block needs more thermal (heat) energy to heat it up by 10°C and which block loses more thermal (heat) energy as it cools back to room temperature?

	more energy	
	heating	cooling
A	X	X
B	X	Y
C	Y	X
D	Y	Y

Question 15

A solid is heated from room temperature. The graph shows how its temperature changes with time as it is heated constantly.



Between which labelled points on the graph is the substance partly solid and partly liquid?

- A between P and Q
- B between Q and R
- C between R and S
- D between S and T

Question 16

A circular metal disc is heated.

Which quantity decreases?

- A its density
- B its diameter
- C its thickness
- D its volume

Question 17

The same quantity of thermal (heat) energy is given to two objects X and Y. The temperature rise of object X is less than the temperature rise of object Y.

What accounts for this difference?

- A X has a larger thermal capacity than Y.
- B X is a better thermal conductor than Y.
- C Y has a larger thermal capacity than X.
- D Y is a better thermal conductor than X.

Question 18

A block of copper and a block of lead are heated. The internal energy of each block increases by the same amount.

The block of copper has a lower thermal capacity than the block of lead.

Which conclusion can be made from this information?

- A. The temperature increase of the copper is greater than the temperature increase of the lead.
- B. The temperature increase of the copper is the same as the temperature increase of the lead.
- C. The temperature increase of the copper is less than the temperature increase of the lead.
- D. The melting point of copper is lower than the melting point of lead.

Question 19

The diagram shows a mercury-in-glass thermometer. The scale of the thermometer has not been marked.



The length l increases uniformly with temperature.

The length l is measured when the thermometer bulb is placed in water at $0\text{ }^{\circ}\text{C}$, and also when it is in water at $100\text{ }^{\circ}\text{C}$. The table shows the results.

temperature/ $^{\circ}\text{C}$	length l /cm
0	2.0
100	26.0

What is the value of l when the bulb is placed in water at $50\text{ }^{\circ}\text{C}$?

- A 12.0 cm B 13.0 cm C 14.0 cm D 16.0 cm

Thermal Properties & Temperature

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Thermal Properties & Temperature
Booklet	Question Paper 2

Time allowed: 23 minutes

Score: /18

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

When steam condenses it becomes liquid water. When liquid water solidifies it becomes ice.

What happens to the temperature of steam while it is condensing, and what happens to the temperature of water while it is solidifying?

	temperature of steam while it is condensing	temperature of water while it is solidifying
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

Question 2

A thermometer has graduations which start at -10°C and end at 110°C .



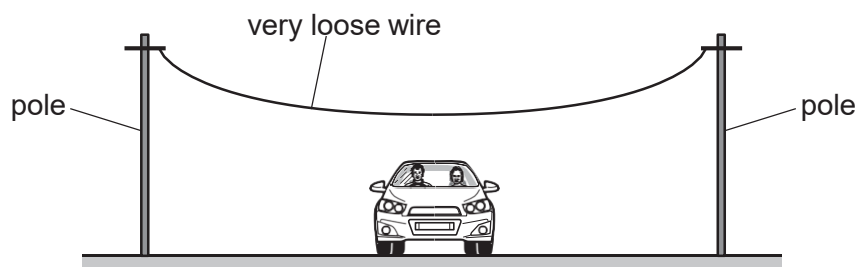
What is the lower fixed point and what is the upper fixed point of the Celsius scale?

	lower fixed point / $^{\circ}\text{C}$	upper fixed point / $^{\circ}\text{C}$
A	-10	100
B	-10	110
C	0	100
D	0	110

Question 3

A telephone engineer connects a wire between two poles when the weather is very cold.

He makes the wire very loose. The wire passes over a road.



The weather changes and it becomes very hot.

What could happen to the wire and why?

	what could happen	why
A	it breaks	it contracts
B	it breaks	it expands
C	it sags and touches cars on the road	it contracts
D	it sags and touches cars on the road	it expands

Question 4

In an experiment, a thermometer is placed in a test-tube of hot liquid. The temperature of the liquid is recorded every half minute. The table shows the results.

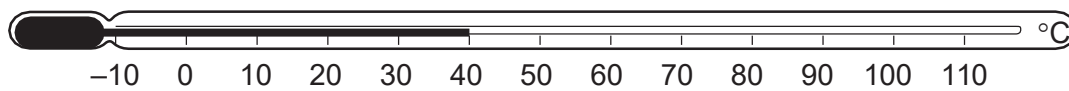
time/minutes	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5
temperature/°C	73	65	59	55	55	55	51	48	45	42	40	38	36	35	34	33

What is the melting point of the substance?

- A 0°C B 33°C C 55°C D 73°C

Question 5

A liquid-in-glass thermometer is marked with a scale in $^{\circ}\text{C}$.



What is the temperature difference between the two fixed points for this thermometer?

- A 40 $^{\circ}\text{C}$ B 50 $^{\circ}\text{C}$ C 100 $^{\circ}\text{C}$ D 120 $^{\circ}\text{C}$

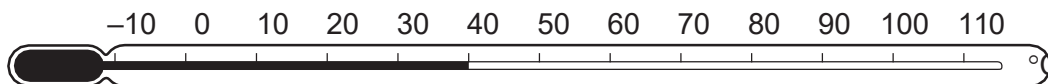
Question 6

Which statement gives the thermal capacity of a solid body?

- A. the energy needed to melt the body without a change in temperature
- B. the energy per degree Celsius needed to raise the temperature of the body
- C. the increase in the volume of the body when its temperature is raised by one degree Celsius
- D. the total amount of internal energy in the body

Question 7

Which points are the fixed points of the liquid-in-glass thermometer shown?



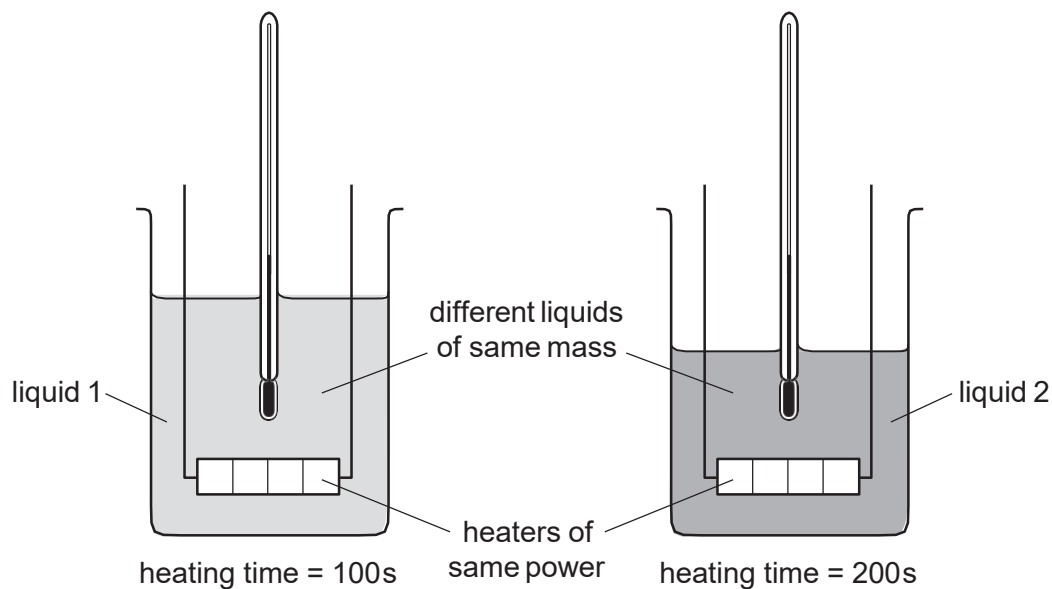
- A. the beginning and end points of the column of liquid
- B. the beginning and end points of the thermometer scale
- C. the points marked 0°C and 100°C
- D. the top and bottom points of the thermometer bulb

Question 8

Equal masses of two different liquids are put into identical beakers.

Liquid 1 is heated for 100 s and liquid 2 is heated for 200 s by heaters of the same power.

The temperature of both liquids increases by the same amount.

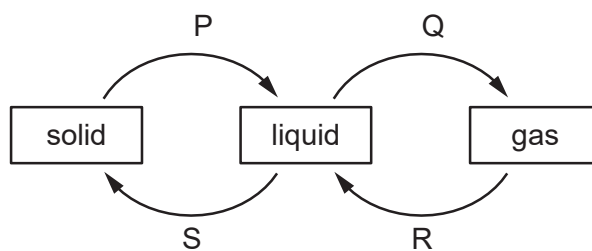


Which statement is correct?

- A. Both liquids receive the same amount of energy.
- B. Liquid 1 receives more energy than liquid 2.
- C. Both liquids have equal thermal capacity.
- D. The thermal capacity of liquid 1 is less than the thermal capacity of liquid 2.

Question 9

The diagram shows four labelled changes of state between solid, liquid and gas.

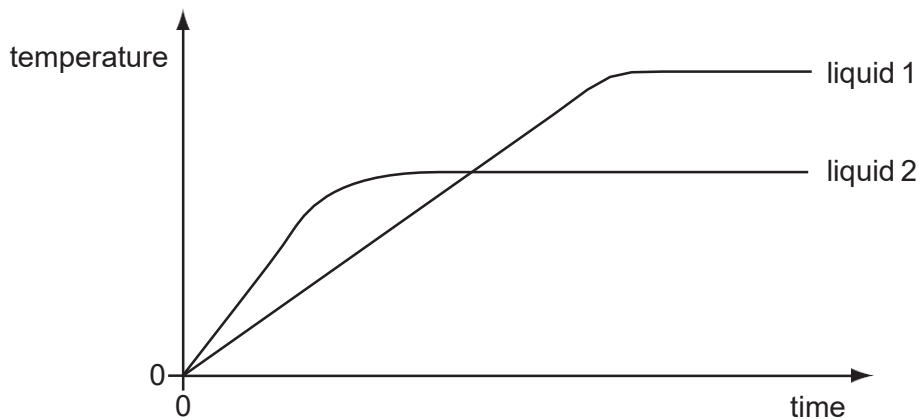


Which changes need an energy input?

- A P and Q B Q and R C R and S D S and P

Question 10

Equal masses of two different liquids are heated using the same heater. The graph shows how the temperature of each liquid changes with time.

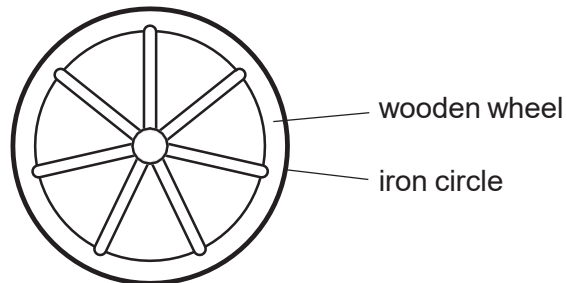


What does the graph tell us about the liquids?

- A Liquid 1 has a higher melting point than liquid 2.
- B Liquid 1 has a higher boiling point than liquid 2.
- C Liquid 1 starts to melt sooner than liquid 2.
- D Liquid 1 starts to boil sooner than liquid 2.

Question 11

A wooden wheel can be strengthened by putting a tight circle of iron around it.



Which action would make it easier to fit the circle over the wood?

- A. cooling the iron circle
- B. heating the iron circle
- C. heating the wooden wheel and cooling the iron circle
- D. heating the wooden wheel but not heating or cooling the iron circle

Question 12

Which pair contains **only** physical quantities that vary with temperature and so could be used in making a thermometer?

- A. activity of a radioactive source, volume of a gas
- B. mass of a liquid, volume of a liquid
- C. activity of a radioactive source, mass of a solid
- D. volume of a gas, volume of a liquid.

Question 13

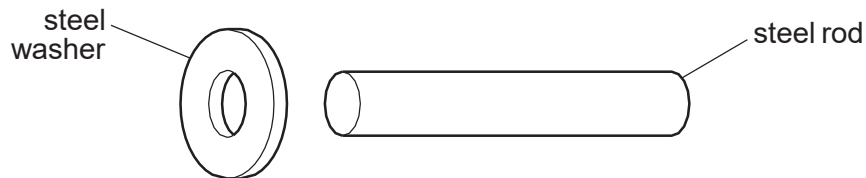
A heater supplies 80 J of energy to a block of metal. The temperature of the block rises by 20°C .

What happens to the block of metal when its temperature falls by 10°C ?

- A Its internal energy decreases by 40 J.
- B Its internal energy decreases by 160J.
- C Its internal energy increases by 40 J.
- D Its internal energy increases by 160J.

Question 14

An engineer wants to fix a steel washer on to a steel rod. The rod is just too big to fit into the hole of the washer.



How can the engineer fit the washer on to the rod?

- A. Cool the washer and put it over the rod.
- B. Cool the washer and rod to the same temperature and push them together.
- C. Heat the rod and then place it in the hole.
- D. Heat the washer and then place it over the rod.

Question 15

An ice cube at a temperature of 0°C is put into a drink at a temperature of 10°C .

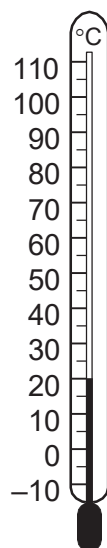
After a short time, some of the ice has melted and the drink has cooled to a temperature of 8°C .

What is the temperature of the remaining ice?

- A 0°C B 2°C C 4°C D 8°C

Question 16

The diagram shows a thermometer calibrated in degrees Celsius.



What are the values of the lower fixed point and of the upper fixed point on the Celsius scale?

	lower fixed point/ $^{\circ}\text{C}$	upper fixed point/ $^{\circ}\text{C}$
A	-10	110
B	0	20
C	0	100
D	20	100

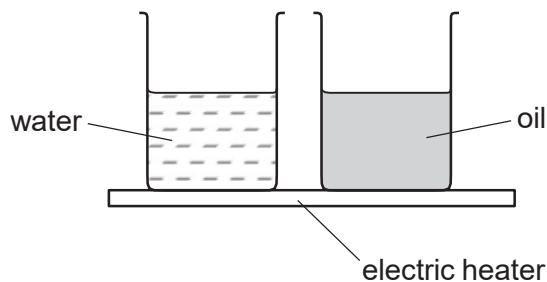
Question 17

To mark the lower fixed point of a Celsius scale on a thermometer, the thermometer should be placed in

- A. pure alcohol.
- B. pure distilled water.
- C. pure melting ice.
- D. pure mercury.

Question 18

The diagram shows an electric heater being used to heat a beaker of water and an identical beaker of oil for several minutes.



The temperature of the water and the temperature of the oil increase constantly. The rise in temperature of the oil is much greater than that of the water.

Why is this?

- A. The oil has a higher boiling point than water.
- B. The oil has a higher thermal capacity than water.
- C. The oil has a lower boiling point than water.
- D. The oil has a lower thermal capacity than water.

Thermal Processes

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Thermal Processes
Booklet	Question Paper 1

Time allowed: 23 minutes

Score: /18

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

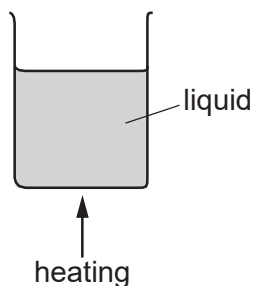
Two otherwise identical cars, one black and one white, are at the same initial temperature. The cars are left in bright sunshine and their temperatures increase. During the night their temperatures decrease.

Which car shows the greater rate of temperature increase and which car shows the greater rate of temperature decrease?

	greater rate of temperature increase	greater rate of temperature decrease
A	black	black
B	black	white
C	white	black
D	white	white

Question 2

A liquid is heated in a beaker.



The density of the liquid changes as its temperature increases. This causes energy to be transferred throughout the liquid.

How does the density change and what is this energy transfer process?

	density	energy transfer process
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection

Question 3

Which processes occur in a metal to cause thermal conduction?

	electron transfer	proton transfer	lattice vibration
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

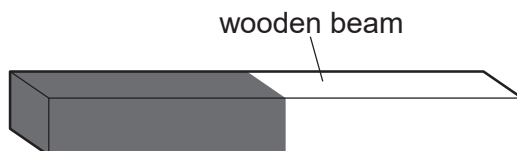
key

✓ = process occurs

X = process does not occur

Question 4

A wooden beam is painted part black and part white. The beam absorbs infra-red radiation from the Sun during the day, and loses infra-red radiation to the surroundings at night.



Which part of the beam heats up more quickly during the day, and which part cools down more quickly at night?

	part heating up more quickly	part cooling down more quickly
A	black	black
B	black	white
C	white	black
D	white	white

Question 5

Which row shows the surface that is the better absorber and the surface that is the better emitter of infra-red radiation?

	better absorber	better emitter
A	black surface	black surface
B	black surface	white surface
C	white surface	black surface
D	white surface	white surface

Question 6

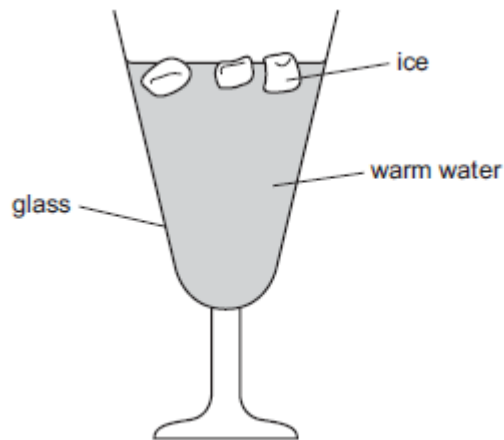
A student suggests some uses for containers made from good thermal conductors and for containers made from poor thermal conductors.

In which row are both suggested uses correct?

	good thermal conductor	poor thermal conductor
A	keeping a cold liquid at a low temperature	transferring thermal energy quickly from a hot liquid
B	keeping a hot liquid at a high temperature	keeping a cold liquid at a low temperature
C	transferring thermal energy quickly from a hot liquid	transferring thermal energy quickly to a cold liquid
D	transferring thermal energy quickly to a cold liquid	keeping a hot liquid at a high temperature

Question 7

The diagram shows some ice being used to lower the temperature of some warm water.



What is the main process by which the water at the bottom of the glass becomes cool?

- A. condensation
- B. conduction
- C. convection
- D. radiation

Question 8

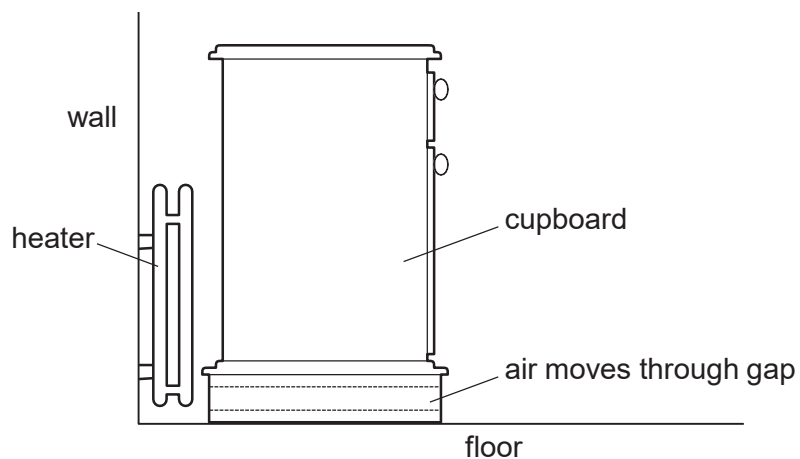
Thermal energy travels through space from the Sun to the Earth. Space is a vacuum.

How is thermal energy transferred from the Sun to the Earth?

- A by conduction only
- B by convection only
- C by radiation only
- D by convection and radiation

Question 9

A cupboard is placed in front of a heater. Air can move through a gap under the cupboard.

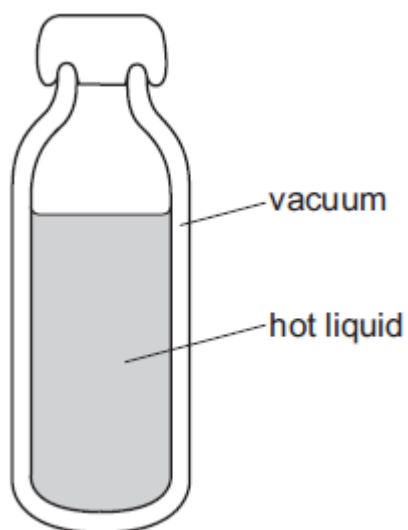


Which row describes the temperature, and the direction of movement, of the air in the gap?

	air temperature	air direction
A	cool	away from the heater
B	cool	towards the heater
C	warm	away from the heater
D	warm	towards the heater

Question 10

The diagram shows a vacuum flask used to keep liquid hot.



How does thermal energy pass through the vacuum?

- A conduction only
- B convection only
- C radiation
- D conduction and convection

Question 11

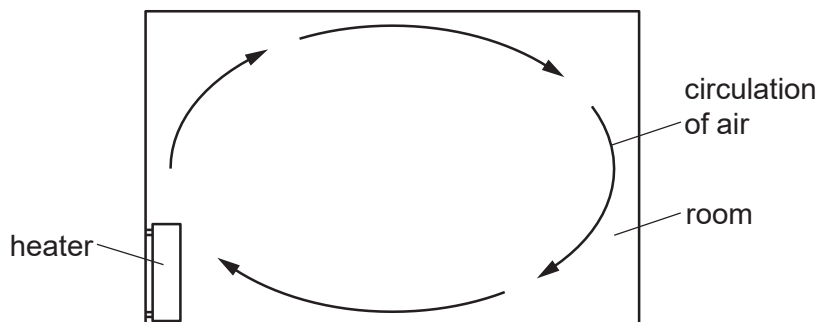
One method of heat transfer involves the energy travelling at a much greater speed than in other methods.

What is the name of this method?

- A conduction
- B convection
- C evaporation
- D radiation

Question 12

The air in a room is heated by a heater. The diagram shows the circulation of the air in the room.



Which statement about the air that is heated is correct?

- A The air contracts and becomes less dense.
- B The air contracts and becomes more dense.
- C The air expands and becomes less dense.
- D The air expands and becomes more dense.

Question 13

Four rods are made from different metals P, Q, R and S. The rods have equal lengths and equal diameters. The rods are heated at one end, in the same way.

The table shows the time taken for the temperature at the other end of each rod to rise by 1.0°C .

Which metal is the best conductor of thermal energy (heat)?

metal	time taken/s
P	35
Q	30
R	45
S	40

A metal P

B metal Q

C metal R

D metal S

Question 14

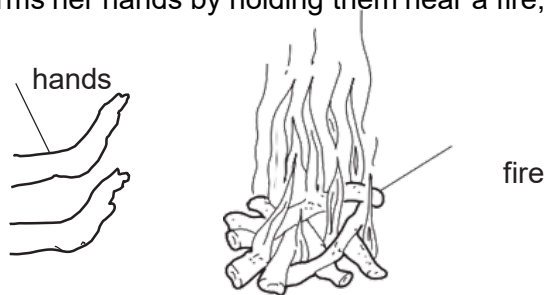
In a refrigerator, the cooling unit can be fitted either at the top or at the bottom. In an oven, the heater can be fitted either at the top or at the bottom.

Which row shows the best position for the cooling unit and the best position for the heater?

	Cooling unit	heater
A	bottom	bottom
B	bottom	top
C	top	bottom
D	top	top

Question 15

A girl is outdoors. She warms her hands by holding them near a fire, as shown.



How does the heat from the fire reach her hands?

- A. conduction only
- B. convection and conduction
- C. convection and radiation
- D. radiation only

Question 16

A heating engineer fits a heater to the ceiling of an office so that workers in the office are kept warm.

How does thermal energy reach the workers below the heater?

- A conduction and convection
- B convection and radiation
- C convection only
- D radiation only

Question 17

Which row shows how heating changes the density of air, and the name of the method of energy transfer caused by this density change?

	density	method of energy transfer
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection

Question 18

What is the name of the process of heat transfer using electromagnetic waves?

- A conduction
- B convection
- C evaporation
- D radiation

Thermal Processes

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Thermal Processes
Booklet	Question Paper 2

Time allowed: 23 minutes

Score: /18

Percentage: /100

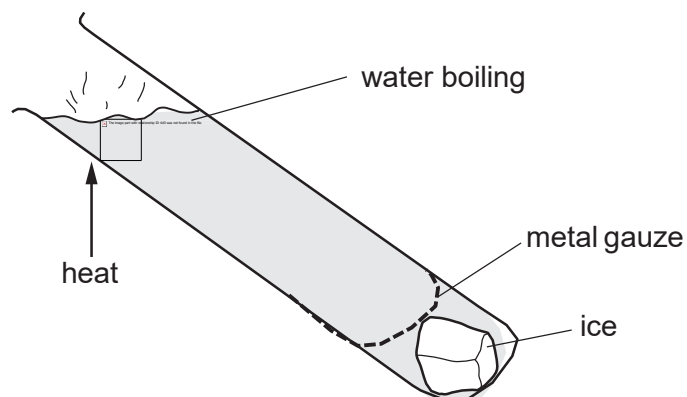
Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

Ice is trapped by a metal gauze at the bottom of a tube containing water.

The water is heated strongly at the top, but the ice only melts very slowly.



Why does the ice melt so slowly?

- A. Heat energy always travels upwards.
- B. Hot water is more dense than cold water.
- C. Metal gauze does not allow heat to pass through.
- D. Water is a poor conductor of heat.

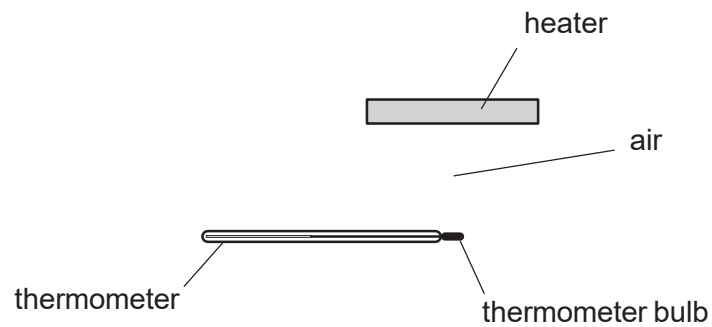
Question 2

Which statement about the transfer of thermal energy is correct?

- A. All metals conduct thermal energy equally well.
- B. Convection can only occur in solids or liquids.
- C. Convection occurs in liquids because hot liquid is more dense than cold liquid.
- D. The radiation that transfers thermal energy is a type of electromagnetic radiation.

Question 3

The diagram shows a heater above a thermometer. The thermometer bulb is in the position shown.

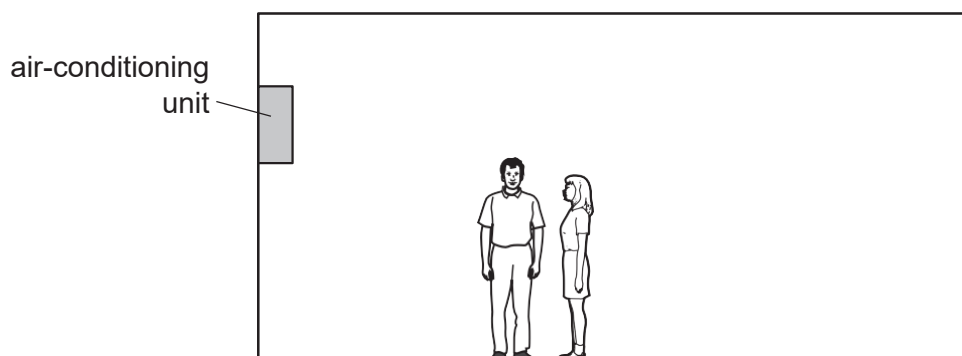


Which row shows how the heat energy from the heater reaches the thermometer bulb?

	conduction	convection	radiation
A	yes	yes	no
B	yes	no	yes
C	no	yes	no
D	no	no	yes

Question 4

The diagram shows an air-conditioning unit on the wall of a room. The unit draws in warm air from the room and releases cold air into the room.



What happens to the cold air and why?

	cold air	why?
A	falls	it is less dense than warm air
B	falls	it is more dense than warm air
C	rises	it is less dense than warm air
D	rises	it is more dense than warm air

Question 5

Which statement about thermal radiation is correct?

- A. It can only occur in a vacuum.
- B. It involves movement of electrons through a material.
- C. It involves movement of atoms.
- D. It is infra-red radiation.

Question 6

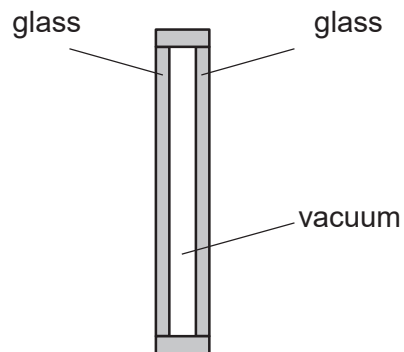
The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

- A The density of the air decreases and the air falls.
- B The density of the air decreases and the air rises.
- C The density of the air increases and the air falls.
- D The density of the air increases and the air rises.

Question 7

One type of double glazing consists of two panes of glass separated by a vacuum.



Which method or methods of energy transfer are prevented by the vacuum?

- A conduction and convection
- B conduction and radiation
- C convection and radiation
- D radiation only.

Question 8

Two methods by which thermal energy can be transferred are conduction and radiation.

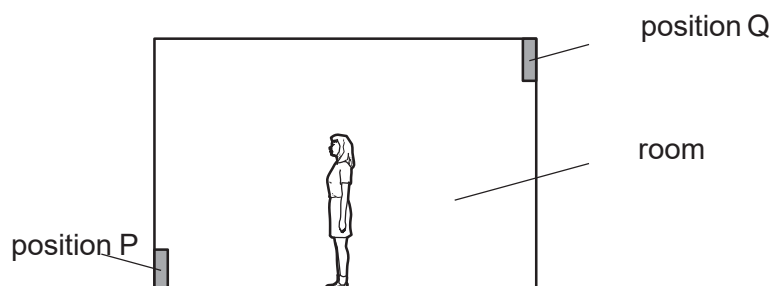
Which statement is correct?

- A. Conduction involves density changes in fluids.
- B. Conduction only occurs in solids.
- C. Radiation cannot occur in a vacuum.
- D. Radiation involves electromagnetic waves.

Question 9

A heater is to be fitted in a room to warm the air throughout the room.

The diagram shows two possible positions to fit the heater, P and Q.

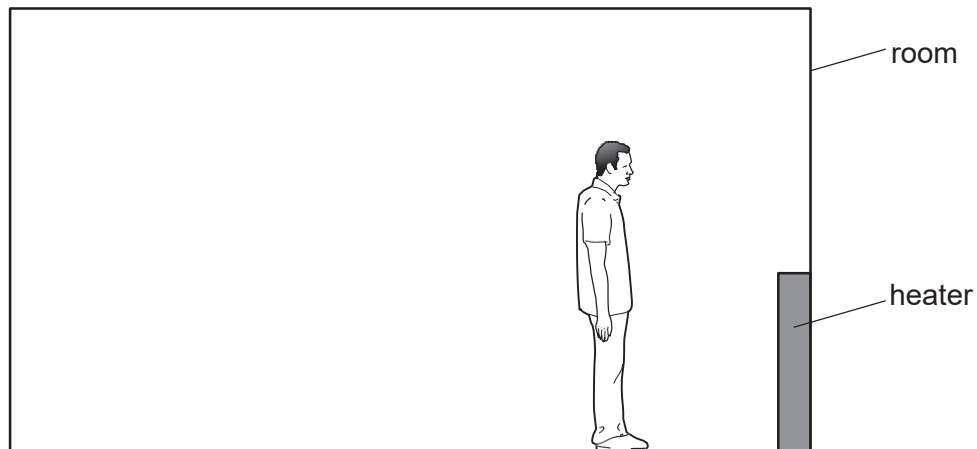


Which position is better and why?

	position	why?
A	P	warmer air is less dense and rises
B	P	warmer air is more dense and rises
C	Q	warmer air is less dense and falls
D	Q	warmer air is more dense and falls

Question 10

A man goes into a cold room and switches on a heater. The man then stands one metre away from the heater. He feels warmer almost immediately.



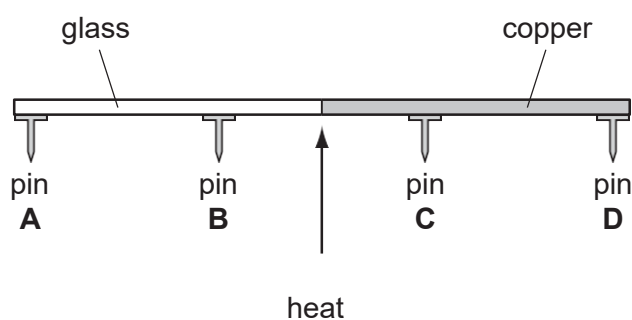
How is thermal energy transferred from the heater to the man so quickly?

- A. by conduction, convection and radiation
- B. by conduction only
- C. by convection only
- D. by radiation only

Question 11

A rod is made half of glass and half of copper. Four pins, A, B, C and D are attached to the rod by wax. The rod is heated in the centre as shown.

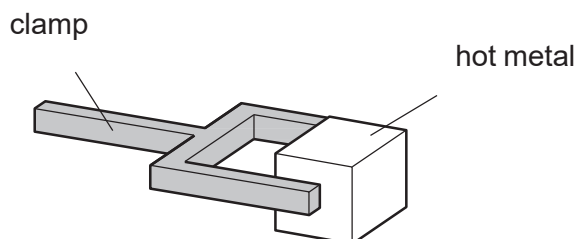
Which pin falls off first?



Question 12

A piece of hot metal is held by a clamp in a cold room. The air next to the metal becomes hot.

The density of the air changes and the air moves.



Which row shows the density change of the air and the direction in which the air moves?

	density of air	movement of air
A	decreases	downwards
B	decreases	upwards
C	increases	downwards
D	increases	upwards

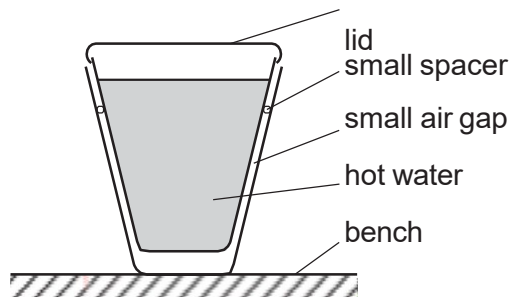
Question 13

Why does convection take place in a liquid when it is heated?

- A. Liquids expand when they are heated.
- B. Liquids start to bubble when they get close to boiling point.
- C. Molecules in the liquid expand when they are heated.
- D. Molecules near to the surface of the liquid escape into the air.

Question 14

Two plastic cups are placed one inside the other. Hot water is poured into the inner cup and a lid is put on top, as shown.



Which statement is correct?

- A. Heat loss by radiation is prevented by the small air gap.
- B. No heat passes through the sides of either cup.
- C. The bench is heated by convection from the bottom of the outer cup.
- D. The lid is used to reduce heat loss by convection.

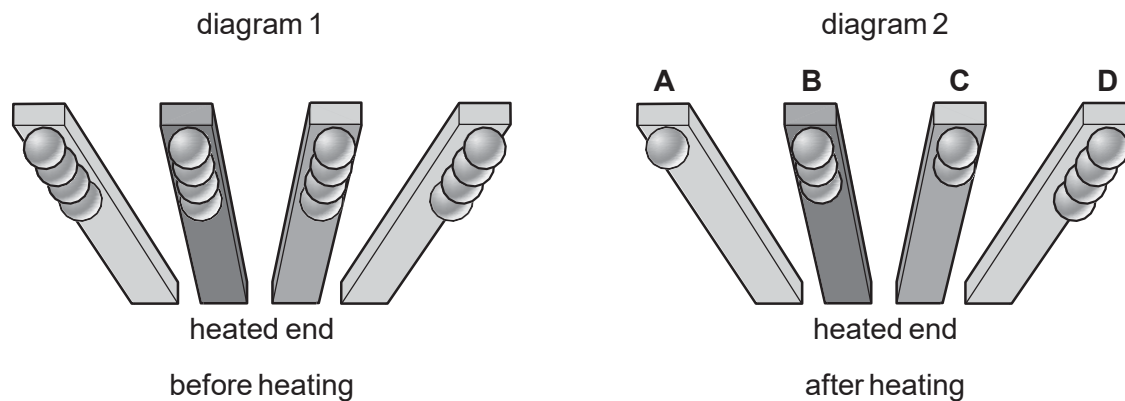
Question 15

An experiment is set up to find out which metal is the best conductor of heat.

Balls are stuck with wax to rods made from different metals, as shown in diagram 1.

The rods are heated at one end. Some of the balls fall off, leaving some as shown in diagram 2.

Which labelled metal is the best conductor of heat?



Question 16

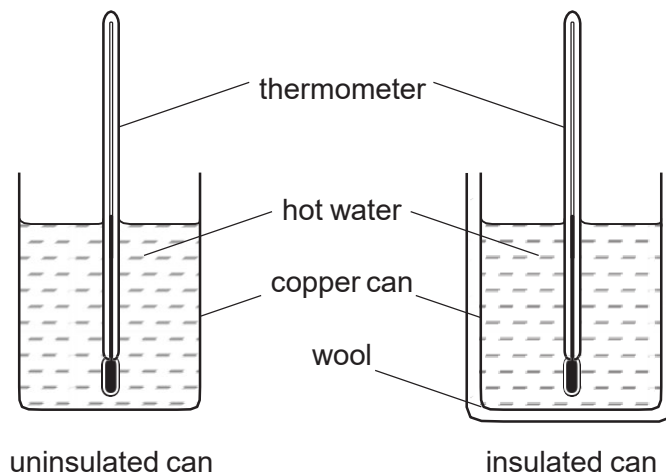
Food is kept in a cool-box which uses two ice packs to keep it cool.

Where should the ice packs be placed to keep all the food as cool as possible?

- A. both at the bottom of the box
- B. both at the top of the box
- C. one at the front and one at the back of the box
- D. one on the left and one on the right of the box

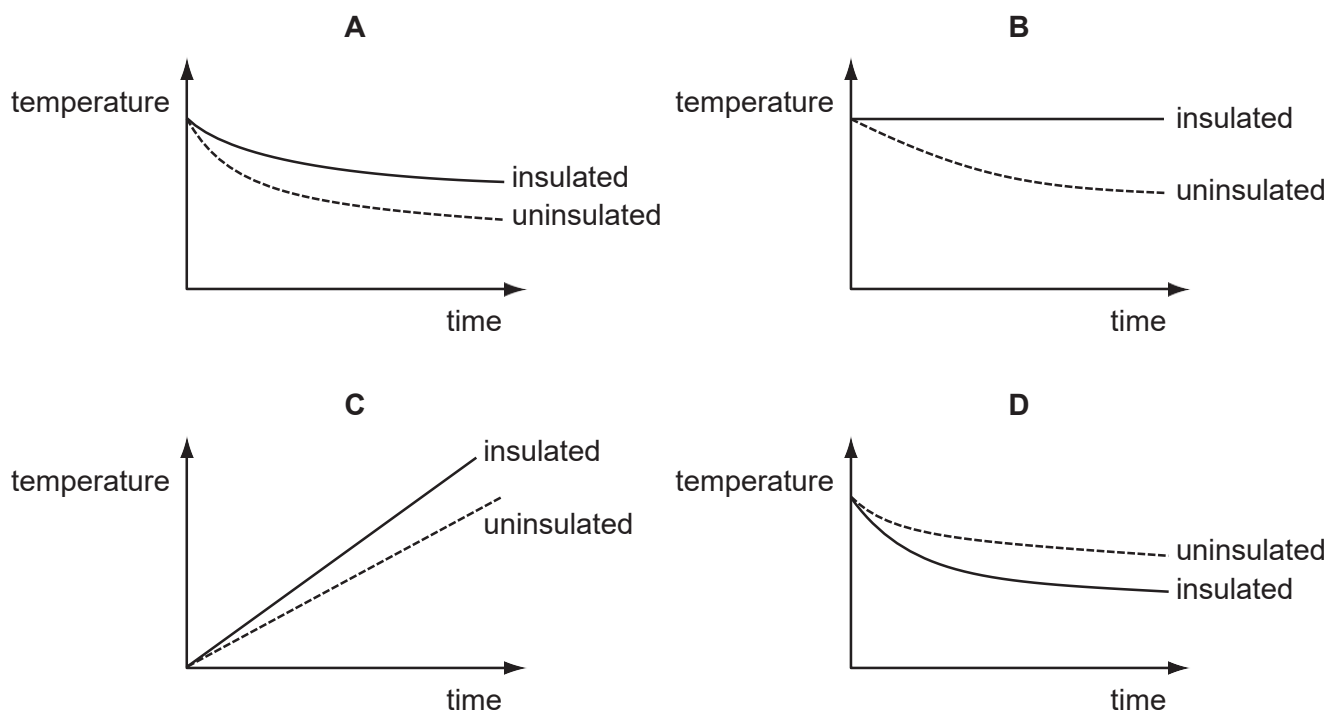
Question 17

Two identical copper cans are filled with boiling water.



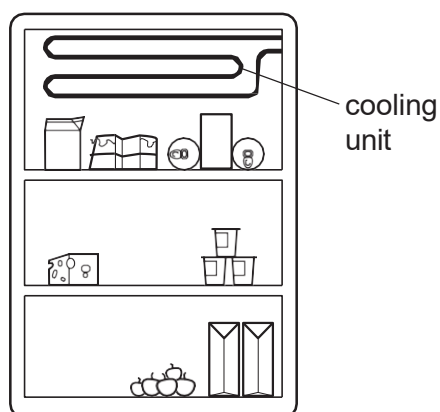
One can is insulated with wool. The temperature of the water in each can is taken every minute for several minutes. Graphs of the results are plotted.

Which graph shows the results obtained?



Question 18

The diagram shows a cooling unit in a refrigerator.



Why is the cooling unit placed at the top?

- A. Cold air falls and warm air is displaced upwards.
- B. Cold air is a bad conductor so heat is not conducted into the refrigerator.
- C. Cold air is a good conductor so heat is conducted out of the refrigerator.
- D. Cold air remains at the top and so prevents convection.